Online Modelling for Real-Time Chiller Efficiency Optimization: A Verified Co-Simulation Testbed with Load Forecasting, Online Efficiency Tracking and Staging-Decision Analysis

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Abstract. Chiller plants account for a large share of commercial-building electricity, yet most run on fixed sequencing rules tuned at commissioning. We describe a reproducible co-simulation testbed (3R2C building, three-chiller water-cooled plant with DOE-2 "Electric EIR" curves, time-of-use tariff) with a verified conventional baseline, and add a learning layer to it: a walk-forward-evaluated gradient-boosting load forecaster, and recursive-least-squares (RLS) estimators that track efficiency drift online. A quasi-static decision analysis then asks how much each kind of model error matters for the discrete choice of how many chillers to run. On synthetic data, forecasting reaches CV(RMSE) of 2.3 %, 3.3 % and 5.3 % at 15 min, 1 h and 4 h; online RLS halves one-step power-prediction error under injected drift (2.8 % vs. 5.5 %). Staging analysis gives a more sober message: errors in model *scale* (efficiency level ±30 %) and realistic forecast errors change the stage-count decision by well under 0.1 % of plant energy, whereas errors in the part-load curve *shape* cost up to 2.8 %. The dominant avoidable loss is the sequencing rule itself (1.4–3.3 %), and it errs in one direction only—it runs too many chillers on 35.9 % of steps and too few on none—so part of that loss is recoverable by retuning one threshold, with no learning layer at all. All results come from simulation with placeholder tariffs; code is provided for exact reproduction.

Keywords: chiller plant, online identification, recursive least squares, load forecasting, model predictive control, co-simulation

1 Introduction

Heating, ventilation and air-conditioning (HVAC) is among the largest electrical end uses in commercial buildings, and in tropical climates the chilled-water plant dominates it. Chiller performance is not static: condenser fouling, refrigerant charge, wear and changing load patterns move the efficiency surface away from its commissioning state, while staging decisions—how many chillers to run—are

still commonly made by threshold rules. Model predictive control (MPC) with forecasts is the natural remedy [1–3], but its benefit depends on two things that are rarely quantified together: how well the load and the plant efficiency can be predicted, and how much an error in either actually changes the operating decision.

This paper takes a deliberately measurement-first approach. Before any predictive controller is built, we (i) construct a testbed whose numerical correctness and conventional baseline are verified, (ii) add the learning components that MPC would rely on—a load forecaster and an online efficiency tracker—and (iii) quantify, by decision analysis, how their errors propagate into the staging decision. The aim is to establish *where* a learning layer can pay off before committing to a controller. Our contributions are:

1. an open, verified Python co-simulation testbed with energy-balance, integrator, time-step and rating-standard checks, and a sensitivity analysis showing that the strategy comparison is more robust than absolute consumption (Section 3);
2. a walk-forward evaluation of short-horizon cooling-load forecasting, including the value of an (imperfect) weather forecast (Section 4);
3. a gray-box RLS estimator of chiller efficiency, in scalar and three-parameter forms, evaluated under two injected drift scenarios (Section 5);
4. a decision analysis separating errors in efficiency *scale*, in part-load *shape* and in load forecasts, with the finding that the first matters little and the second a great deal (Section 6).

Relation to the wider research programme. This paper reports one part of a larger programme on online modelling and predictive control of chiller plants, whose end goal is a closed-loop MPC with verified efficiency gains. What is reported here is the enabling layer (testbed, forecasting, online identification, decision analysis); the controller itself, its ablation against a static efficiency model, and any claim of achieved savings are *not* part of this paper. The 1.43 % figure below is the headroom of better staging on a simulated plant, not a demonstrated saving.

We state limits plainly: all data are simulated, the tariff is a placeholder, the decision analysis is quasi-static, and no closed-loop MPC is reported here (Section 7).

2 Related Work

Chiller modelling. Regression-based models of the DOE-2 type express electrical input as the product of a rated value and curves in load fraction and temperatures; they are standard in EnergyPlus [4] and have been studied for calibration from plant data [5, 6]. Ratings follow AHRI 550/590 [7]. *Chiller-plant optimisation* has used simplified adaptive models with evolutionary search [8], quadratic plant representations [9] and practical sequencing guidance [10]. *Predictive control* of buildings with weather forecasts is reviewed in [1]; field trials report

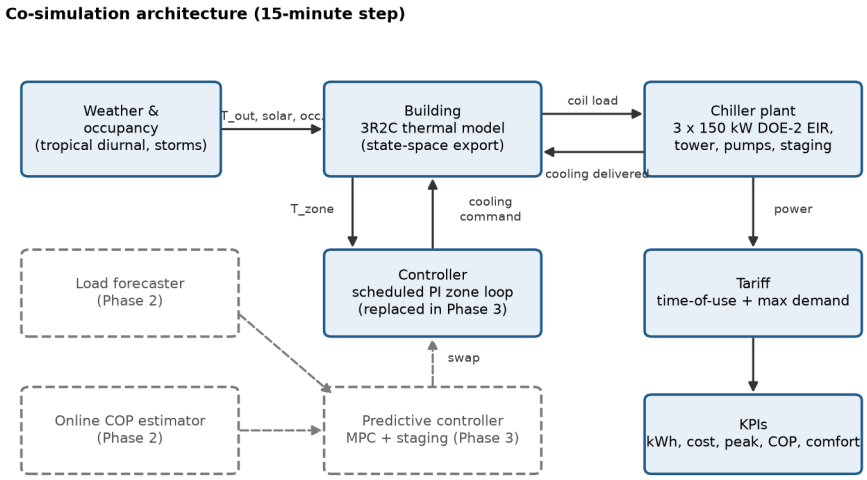


Fig. 1. Co-simulation architecture. Solid blocks are implemented; the learning layer (Sections 4–5) attaches to the same interfaces.

cost benefit [3], and learning-based control is surveyed in [11]. *Load prediction.* Data-driven building energy prediction is reviewed in [12]; gradient-boosted trees [13] remain highly competitive on tabular inputs [14], and time-series evaluation needs forward-chaining rather than random splits [15]. *Online identification.* RLS with forgetting is a classical tool for tracking slowly varying parameters [16–18]; gray-box building models are discussed in [19, 20]. To our knowledge, what is less often reported is the separation we make here between error in efficiency level, in part-load shape, and in load forecast, each measured in units of staging-decision loss. This statement about the literature has not yet been established by a systematic review and should be read as a hypothesis motivating the study.

3 Testbed and Baseline

3.1 Architecture

Weather, building, plant, tariff and controller are separate modules with fixed interfaces, coupled at a 15-minute step (Fig. 1). The controller is the only block that changes between conventional control and the learning-based variants.

Building. A 3R2C lumped thermal model of a $\approx$ 5,000 m^2 office (two thermal modes, time constants 0.85 h and 10.4 h), driven by a synthetic tropical weather generator with cloud variability and afternoon storms. The model exports its continuous-time state-space matrices, so a predictive controller can use the same dynamics without reformulation. Latent load is a fixed multiplier (1.3) on sensible load. Peak coil load is about 361 kW.

Plant. Three 150 kW constant-speed water-cooled centrifugal chillers use the DOE-2 form

$$P_{ch} = n\,\frac{C(T_{cw})}{\mathrm{COP}_{ref}}\;\mathrm{EIRfT}(T_{chw}, T_{cw})\;\mathrm{EIRfPLR}(\mathrm{PLR}),$$
$$\mathrm{EIRfPLR} = a + b\,\mathrm{PLR} + c\,\mathrm{PLR}^2, \tag{1}$$

with n running units sharing the load, $a + b + c = 1$ (nominal $a, b, c = 0.30, 0.23, 0.47$), and on/off cycling below 15 % part-load. A cooling tower (approach 3–5 K over wet-bulb), variable-speed chilled-water pump, per-unit condenser pumps and tower fans complete the plant. Curve coefficients are *synthetic*, not manufacturer data.

Conventional control. The baseline is a scheduled plant with a tuned PI zone loop with anti-windup, threshold chiller sequencing (stage up above 90 % of unit capacity, down below 75 % of the reduced stage) and a fixed supply temperature—deliberately competent, since savings are only meaningful against a credible reference. A second reference, fixed-schedule pre-cooling, is included as an intermediate benchmark. The PI output limit is set to 400 kW sensible (520 kW at the coil) so that it exceeds what the plant can deliver and never binds in either strategy; Section 7 reports what happens when it does.

3.2 Verification and Baseline Results

Verification (does the code solve the equations correctly?): against the closed-form matrix-exponential solution of the linear thermal model the RK4 integrator deviates by at most 7.9×10^{-5} K over 24 h; first-law closure over 14 days leaves 2.1×10^{-5} % of throughput; and the 15-minute step lies within 0.18 % in energy and 0.03 % in peak of a 5-minute solution (a 60-minute step would overstate energy by 4.6 % and understate peak by 5.1 %). **Plausibility** (not validation): the plant rates at 0.597 kW/ton full-load and 0.556 kW/ton IPLV under AHRI 550/590 conditions, and five plant-level metrics (seasonal COP 4.93, peak cooling density 72 W/m^2, auxiliary share 17.7 %, tower approach 3.9 K, daily plant electricity) fall inside indicative ranges. **Empirical validation against measured plant data has not been done**; the ASHRAE Guideline 14 statistics [22] (NMBE, CV(RMSE)) are implemented but unrun for lack of data.

Table 1 gives 14-day results (two spin-up days discarded). Rule-based pre-cooling is *worse* than the baseline on cost by 9.0 %: it saves RM 22 of energy charge but creates a new 04:30 demand peak (94.2 vs. 72.2 kW, adding RM 311 of demand charge) and uses 4.3 % more energy. This negative result separates "prediction helps" from the weaker "any load shifting helps". Under ±25 % perturbation of eight building parameters, absolute energy moved by −25 to +26 % (the model is uncalibrated), but the energy gap between the two strategies stayed within 3.9–5.9 % (nominal 4.3 %) in all 16 cases, because parameter error is largely common to both strategies. The cost gap (0.1–14.4 %) is less robust because the demand charge hinges on a single interval; costs here use placeholder RM rates and indicate direction only.

Table 1. Baseline and rule-based pre-cooling, 14 simulated days (placeholder tariff).

Strategy	Energy (kWh)	Cost (RM)	Peak (kW)	Plant COP
Schedule + PI	7,665	3,203	72.2	4.93
Pre-cooling	7,992	3,492	94.2	4.97

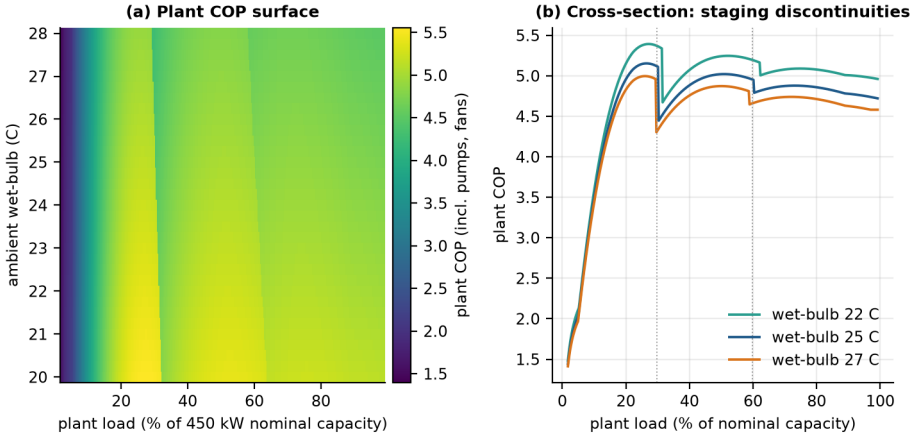


Fig. 2. Plant COP over the operating envelope; staging produces sharp discontinuities.

3.3 Non-Convex Efficiency Surface

Mapping plant COP over load and wet-bulb temperature (Fig. 2) shows that efficiency rises to a local peak just below each stage-up threshold and then drops as the next unit starts: by 12 % at the first threshold (COP 5.13 to 4.51 at 25 °C wet-bulb) and 3 % at the second. The surface depends weakly on wet-bulb compared with load. The drops depend on the assumed part-load curve and auxiliary power and are therefore properties of this synthetic plant. The surface implies a discrete decision that a continuous model cannot represent—which motivates Section 6.

4 Cooling-Load Forecasting

Setup. The testbed was run for 120 days under the baseline strategy; the forecasting target is the coil load H steps ahead ($H \in \{1, 4, 16\}$, i.e. 15 min, 1 h, 4 h), evaluated during scheduled hours (06:00–18:30), about 4,600 points per horizon. Inputs are time of day (sine/cosine), occupancy, current and lagged loads (t, $t-15$, $t-30$, $t-60$ min), zone temperature, current weather, the load at the same time the previous day, and a *noisy weather forecast* for the target time (dry-bulb, solar and humidity with error standard deviations that grow with horizon: 0.3/0.6/1.2 K, 5/10/25 % and 1.5/3/6 % RH). Models: persistence,

Table 2. Walk-forward load-forecast accuracy, CV(RMSE) in %, noisy weather forecast.

Model	15 min	1 h	4 h
Persistence	20.4	30.5	51.6
Seasonal naive	10.0	10.0	10.0
Ridge	3.3	6.6	8.3
Gradient boosting (HGB)	**2.3**	**3.3**	**5.3**

seasonal-naive (yesterday), ridge regression and histogram gradient boosting (HGB, scikit-learn [21]). Evaluation is *walk-forward*: initial training on 28 days, then 14 successive 7-day test folds with an expanding training window, never training on the future. Accuracy is CV(RMSE)$= 100\sqrt{\overline{e^2}}/\bar{y}$. Errors e are pooled over all test points (no degrees-of-freedom correction), and $\bar{y}$ is the mean observed coil load over the same points (≈ 244 kW, scheduled hours only); the figure is therefore not identical to the hourly CV(RMSE) of Guideline 14 [22], which is defined for calibration, not forecasting. The metric was recomputed independently of the reporting code, and the training window of every fold ends before the first test target.

Results. Table 2 and Fig. 3 show HGB beating every baseline at every horizon, with a skill of about 0.89 over persistence in RMSE; errors rise with horizon but stay within 5.3 % at 4 h and are stable across folds (4 h: 3.3–6.9 %). A weather forecast helps mainly at longer horizons: at 4 h CV(RMSE) is 4.1 % with a perfect forecast, 5.3 % with the noisy one and 5.8 % without; at 15 min it is irrelevant (2.2–2.3 %). Because the load here is generated by a smooth deterministic model plus synthetic weather, these numbers are *optimistic* compared to a real building with irregular occupancy; they show that the pipeline works and how weather-forecast quality enters, not what accuracy to expect on site. A recurrent network was not trained as a comparison.

5 Online Efficiency Tracking

Gray-box estimators. Measured chiller electrical power is modelled on the commissioning structure (1) with a slowly varying multiplier k (relative COP^{-1}): $y_t = k\,\phi_t$, where ϕ_t is the commissioning-model power at the observed load, stage count and condenser water temperature, and y_t is sub-metered power with 2 % multiplicative noise. The recursive update with forgetting factor λ is

$$K_t = \frac{P_{t-1}\phi_t}{\lambda + \phi_t^2 P_{t-1}}, \quad \hat{k}_t = \hat{k}_{t-1} + K_t(y_t - \hat{k}_{t-1}\phi_t), \quad P_t = \frac{P_{t-1} - K_t\phi_t P_{t-1}}{\lambda}. \tag{2}$$

Because (1) is linear in (a, b, c), a three-parameter estimator with vector regressor $\boldsymbol{\phi}_t = \frac{nC}{\mathrm{COP}_{ref}}\mathrm{EIRfT}\,[1, \mathrm{PLR}, \mathrm{PLR}^2]$ follows the same recursion in vector form; samples in the cycling region ($\mathrm{PLR} < 0.15$) are excluded.

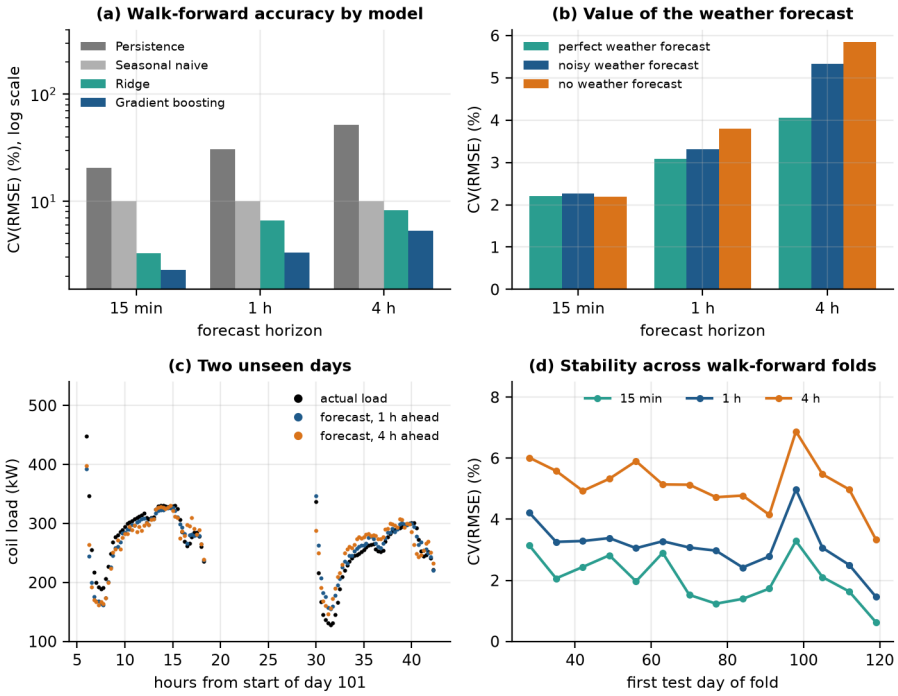


Fig. 3. Forecasting. (a) accuracy by model and horizon; (b) value of the weather forecast; (c) two unseen days; (d) CV(RMSE) per walk-forward fold.

Drift scenarios (injected). (A) *Fouling and cleaning*: efficiency degrades linearly by 10 % from day 15 to 60, is restored to −1 % by a cleaning at day 60, then degrades 6 % over 45 days. (B) *Part-load flattening*: the fixed-loss term a rises from 0.30 to 0.42 over days 20–80 (b, c rescaled so the curve equals 1 at full load). A no-drift control run checks that the estimator does not invent drift (mean $|\hat{k} - 1| = 0.05\,\%$).

Results. The metric is the one-step-ahead power prediction error (before the update) against the static commissioning model, which has error 5.45 % in scenario A (7–9 % at the end of fouling episodes; Fig. 4b). With $\lambda = 0.998$ the RLS error is 2.76 % (mean parameter error 1.3 %); with $\lambda = 0.99$ it is 2.26 % (0.44 %); with $\lambda = 1$ (no forgetting) 3.38 % (2.2 %), since it cannot follow the cleaning event. Meter noise of 2 % sets a floor, and the parameter error is nearly insensitive to noise up to 5 % (Fig. 4d). Shorter memory tracks faster at the cost of noise sensitivity (0.43 % → 0.53 % mean error as noise rises 0.5 % → 5 % at $\lambda = 0.99$), which is the expected bias–variance trade-off. In scenario B the scalar k cannot represent a change of *shape*: its error is 3.41 %, against 2.57 % for the three-parameter curve and 6.50 % for the static model. The curve estimator lags

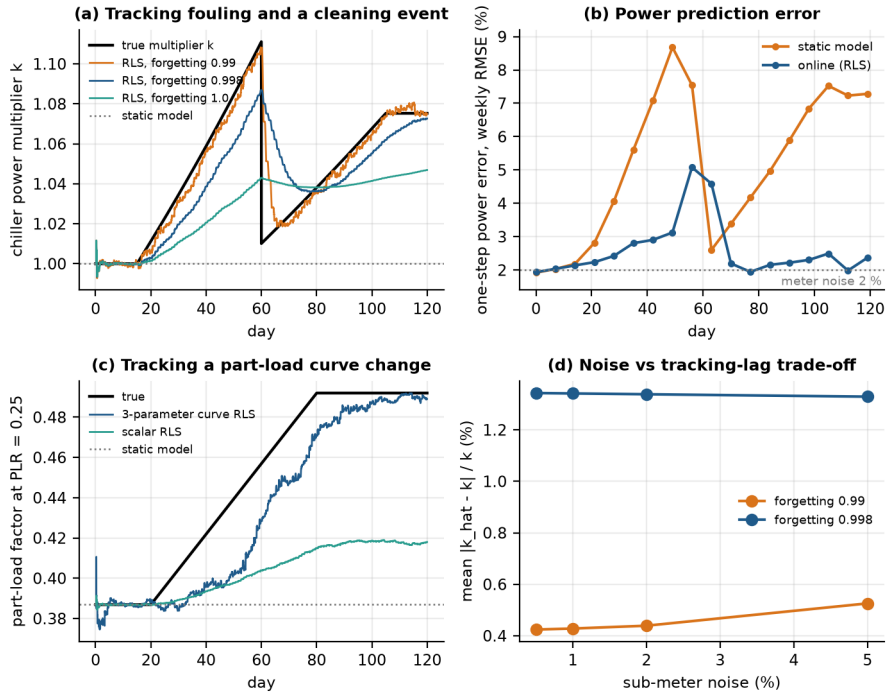


Fig. 4. Online identification. (a) tracking of the multiplier k through fouling and cleaning; (b) weekly power-prediction error; (c) tracking of the part-load factor at PLR $= 0.25$; (d) noise versus forgetting.

the true parameter (Fig. 4c): the regressors $[1, \mathrm{PLR}, \mathrm{PLR}^2]$ are collinear over the operating range, so individual coefficients are weakly identifiable even when predictions are accurate. We therefore report prediction error, and warn that estimated coefficients should not be read as physical quantities.

6 How Much Do Model and Forecast Errors Matter for Staging?

Method. At each scheduled time step with load q and wet-bulb temperature T_{wb}, the quasi-static plant electrical power (chillers, pumps, fans) is evaluated for each feasible stage count $n \in \{1, 2, 3\}$; the *optimal* n^* minimises it under the *true* plant. A decision-maker with an imperfect model (or forecast) picks $\hat{n}$ from its own estimate; we report the *mismatch rate* ($\hat{n} \neq n^*$) and the *excess energy* $\sum P(\hat{n}) / \sum P(n^*) - 1$ evaluated on the true plant. This is a measure of *decision regret*, not of closed-loop savings: it ignores dynamics, ramp constraints and the thermal-storage value of the building, which is where predictive control would add more.

Headroom, and what it is made of. Over 6,000 scheduled steps, the rule-based sequencer chooses a different stage count from the quasi-static optimum on 35.9 % of steps and uses 1.43 % more plant energy (74.4 vs. 73.3 MWh over the run). This is an *upper bound on what better staging alone can recover* on this plant, before any thermal-storage value; the mismatch concentrates in the 20–40 % and 55–70 % load bands (Fig. 5d). Three properties of that 1.43 % matter more than its size.

The rule errs in one direction only. It runs *too many* chillers on 35.9 % of steps and too few on none: the stage-up threshold at 90 % of unit capacity is systematically eager. Part of the headroom is therefore recoverable by retuning a single threshold, with no forecaster and no estimator.

The loss is diffuse and is bought with more switching. It is the sum of many near-ties rather than a few large mistakes—the worst 10 % of steps carry 55 % of it and the median mismatched step wastes 2.8 % of its own power—and reaching the quasi-static optimum would require 532 stage changes over the 120 days against the rule's 377, a 41 % increase that this analysis does not cost.

A structural bound sits behind all of it. The plant runs three chillers on 47.3 % of scheduled steps and two on 45.9 %, at a median part-load ratio of 0.66, and the optimum selects the *minimum feasible* stage count on 98.0 % of steps. On this plant, optimal staging is almost exactly the policy "run as few machines as the load allows", which bounds what any method—learned or not—can win here.

Errors in scale do not matter. Giving the decision-maker a model with the efficiency level wrong by factors of 0.8–1.3 changes the plant energy by at most 0.010 % (mismatch 1–3 %). Likewise under fouling drift (scenario A) a static model loses only 2.5×10^{-4} % and online RLS 3×10^{-5} % versus 1.42 % for the rule-based sequencer. The reason is structural: the chiller multiplier k scales every candidate stage count almost equally, so the ranking of options is unchanged. *Consequently, online tracking of efficiency* level *is valuable for diagnostics, prediction and MPC cost forecasting, but not for choosing the stage count on this plant.* This corrects an expectation we held at the outset—that small model errors flip the staging choice near thresholds.

Errors in shape matter. Errors in the part-load curve parameter a do: with $a = 0.15$ the extra energy is 2.83 % (mismatch 58 %) and with $a = 0.22$ it is 0.98 % (33 %), whereas for $a \geq 0.38$ the results are identical (2.0 % mismatch, 0.005 %) because such a model selects the minimum feasible number of units on 100 % of steps—one saturated regime, not three independent null results. Under part-load drift (scenario B) the static model loses 0.10 %, the scalar RLS 0.13 % (slightly *worse*: a scalar absorbs shape error in the wrong direction), the three-parameter curve RLS 0.001 %, and the rule-based sequencer 3.32 %. Identification therefore pays off for staging only if the estimator can represent the kind of drift that occurs.

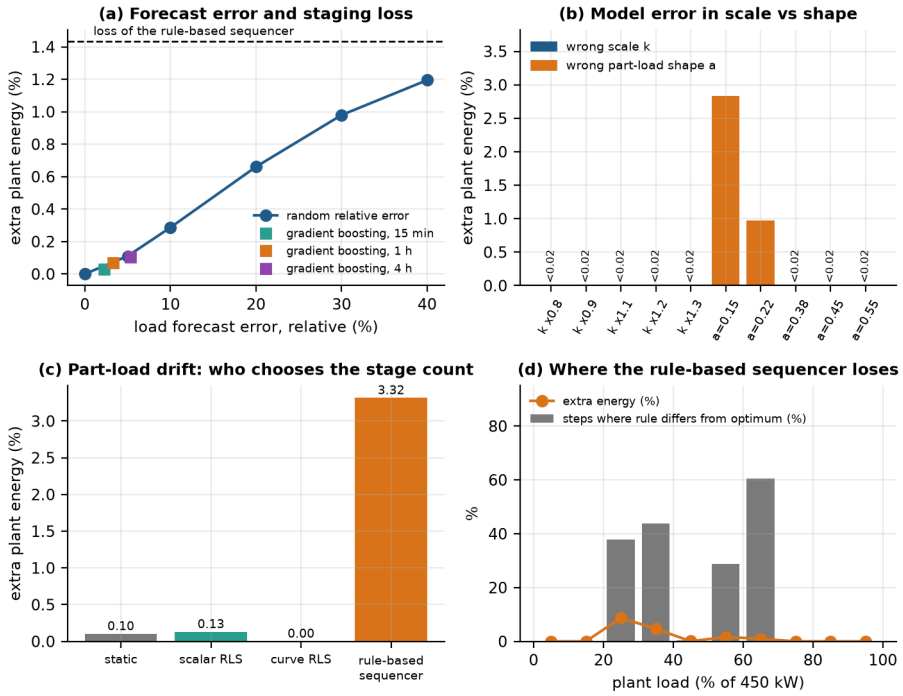


Fig. 5. Decision analysis. (a) staging loss versus load-forecast error (dashed: rule-based sequencer's loss); (b) model error in scale versus shape; (c) who chooses the stage count under part-load drift; (d) where the rule-based sequencer loses, by load band.

Forecast errors. Replacing the true load by a forecast $\hat{q} = q(1+\sigma\varepsilon)$ with random relative error σ yields excess energy 0.11, 0.29, 0.66, 0.98 and 1.20 % at $\sigma = 5, 10, 20, 30, 40$ % (mismatch 6–23 %), i.e. roughly linear in σ and always below the 1.43 % rule-based loss (Fig. 5a). Using the actual HGB forecasts of Section 4, the staging loss is 0.03, 0.07 and 0.10 % for 15 min, 1 h and 4 h horizons (mismatch 1.5, 3.3, 5.0 %). At the accuracy achieved on this synthetic data, forecast error costs one to two orders of magnitude less than the rule-based sequencer's own loss; the accuracy threshold at which a forecast stops being useful for staging is reached only at relative errors well beyond 40 %. Real-building forecasts will be worse than ours, and the threshold should be re-estimated on measured data.

7 Discussion and Limitations

What the evidence supports. Within this testbed: (1) load forecasting to a few per cent CV(RMSE) is achievable and leaves staging regret negligible; (2) online RLS substantially improves power prediction under drift and is cheap; (3) for the discrete staging decision, *scale* information is nearly irrelevant, *shape* information

matters, and the existing rule is itself the dominant loss (1.4–3.3 %). The latter suggests that the main near-term value of a learning layer lies in correcting the sequencing rule and in providing accurate costs to an optimiser, rather than in sharpening an efficiency level.

What it does not support. (i) *All data are simulated*; drift scenarios are injected and the forecaster, the estimators and the plant share the same structural assumptions (the gray-box estimator uses the exact DOE-2 form that generated the data), so structural model error is not tested. (ii) The tariff rates are placeholders; costs indicate direction only, and the baseline's calibration to a real building is absent (Section 3). (iii) The decision analysis is quasi-static: it omits ramping, minimum run times, thermal storage and comfort constraints, and no closed-loop MPC or ablation (static versus online model) is reported. The omission is quantified rather than assumed: the quasi-static optimum needs 41 % more stage changes than the rule (532 against 377), so minimum run times could absorb a large part of the 1.43 %. (iv) Staging losses of 1.4–3.3 % are plant-energy figures from a synthetic constant-speed plant; variable-speed machines flatten the part-load structure and could change the conclusions. (v) No recurrent-network baseline was trained, and a Gaussian-process estimator was not tried. (vi) Only building parameters were perturbed in the sensitivity study; controller parameters were not, and one of them is consequential—see below.

A controller limit that the cost comparison turns on. The PI output saturates at a limit u_{max} that is a tuning constant, not a plant property. An audit of the code found that at the value used in the earlier version of this work (340 kW sensible, i.e. 442 kW at the coil) the pre-cooling strategy saturated at four steps, all at 04:30, and its demand peak—the figure that drives the whole cost penalty—occurred at one of them, while the baseline never saturated. The two strategies were therefore not compared on equal actuator terms. Sweeping the constant moves the cost gap from +2.1 % ($u_{max} = 300$) through +7.3 % (340) to +9.0 % (400 and above, where the limit no longer binds), while the energy gap holds at 4.13–4.27 % throughout. We report the saturation-free case ($u_{max} = 400$), and note the episode as a reason to treat any cost figure from this model as indicative: an unreported controller constant moved it by a factor of four, where the energy comparison was unaffected.

Next steps. Replace synthetic data with measured BMS trends (enabling Guideline 14 calibration); run a structural-mismatch study that generates data with one plant model and identifies with another, since the present estimator is handed the form that produced the data; add minimum run times and start-up penalties and re-derive the headroom; sweep the staging thresholds to see how much of the 1.43 % a retuned rule recovers without any learning; and close the loop with an enumeration-based MPC over the exported state-space model, with the ablation that isolates the online model.

8 Conclusion

We built and verified a chiller-plant co-simulation testbed, added a forecasting and online-identification layer, and measured how its errors translate into staging-decision loss. On synthetic data, forecasting and RLS are accurate and cheap, but the staging decision is robust to errors in efficiency scale and to realistic forecast errors; it is sensitive to the part-load shape, and the existing rule-based sequencer is the largest identified loss. These results define where a learning layer has the most to offer and where it does not, and they set the experimental frame for a predictive controller.

Reproducibility. The full code, README and expected outputs (Phase 1 and 2; one script reproduces all 35 reported key numbers in about one minute; Python 3.13, NumPy, SciPy, scikit-learn, Matplotlib) accompany this paper: https://github.com/mesmail/NLPIR_paper_chiller_efficiency.

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